# Design Patterns

**CS6.401 Software Engineering**

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# Acknowledgements

The materials used in this presentation have been gathered/adapted/generated from various sources as well as based on my own experiences and knowledge

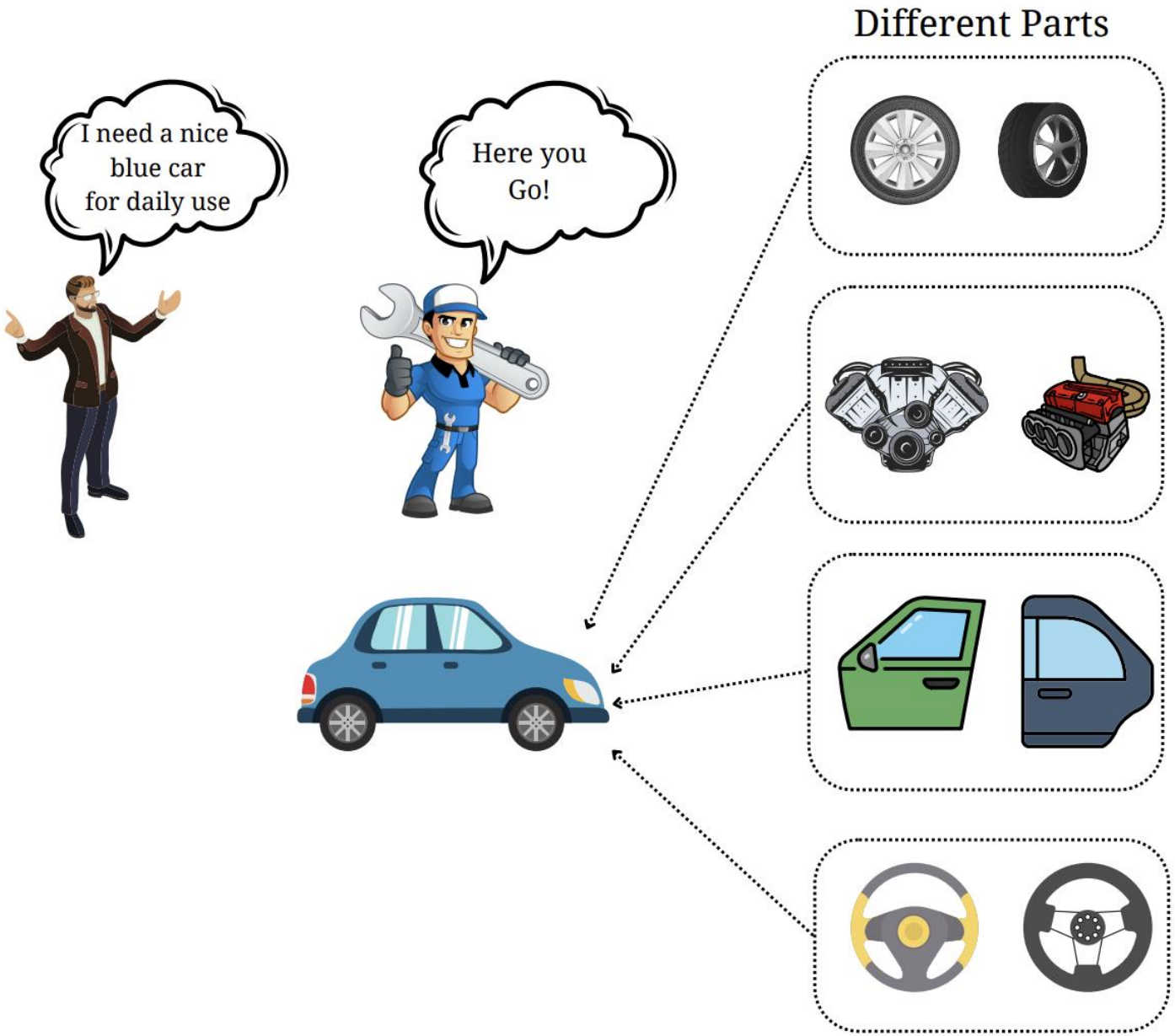
-- Karthik Vaidhyanathan

Sources:

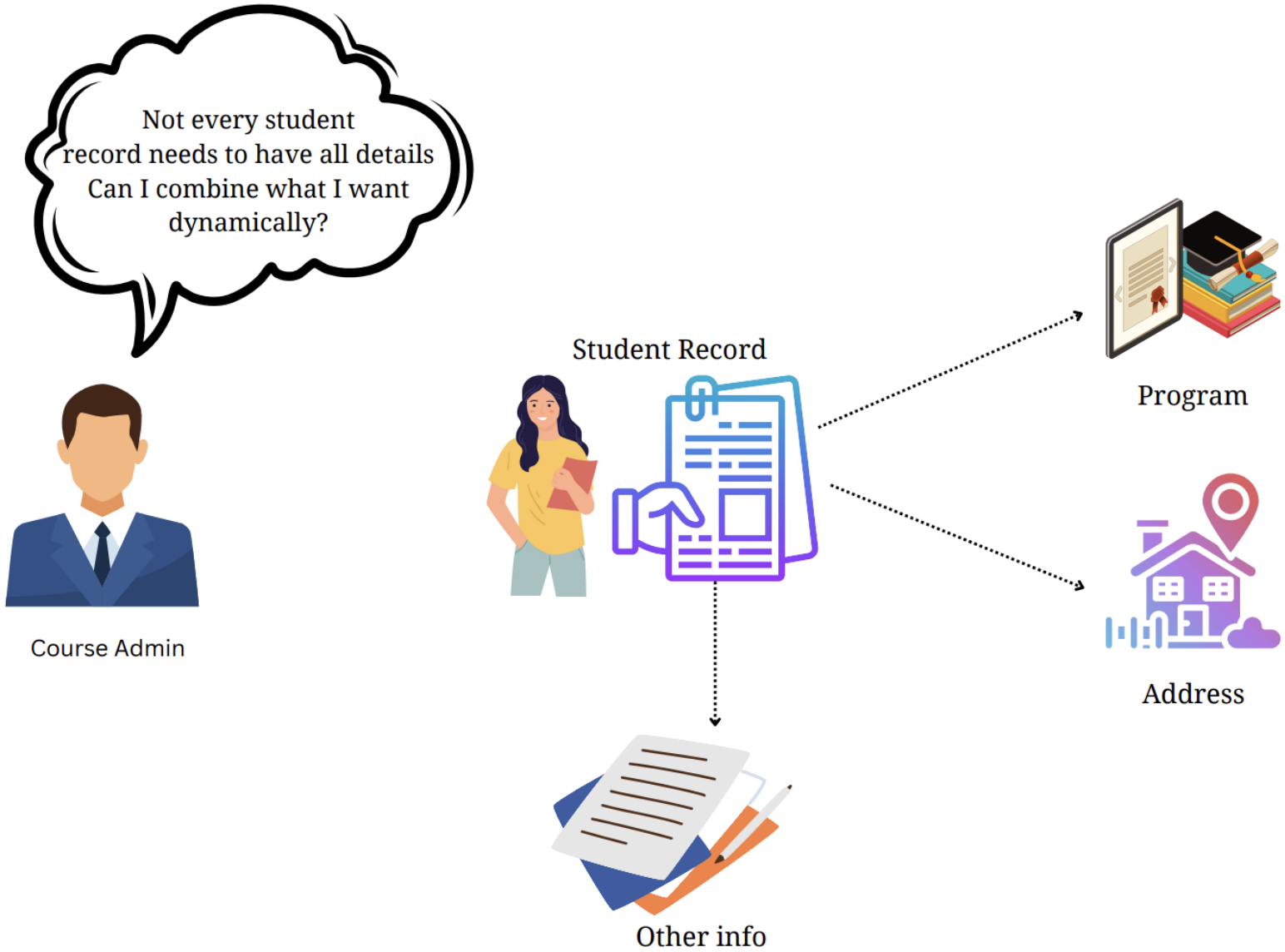
1. **Design Patterns: Elements of Reusable Object-Oriented Software** by Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides
2. **Head first Design Patterns**, Second Edition, Eric Freeman and Elisabeth Robson

# How about building things: Builder Pattern! [Creational]

# Meet the Builder Pattern!



# Meet the Builder Pattern!




How to dynamically build the different types of student records?

# Meet the Builder Pattern

- What if there is a complex object?
- Can we avoid instantiation of a huge constructor?
- Not every time all constructor parameters are required
- Allows extraction of object construction code to separate object
- Creation of an object is just about assembling other objects step by step
- A very decoupled approach to creation

# Builder Pattern: Documentation

**Intent**

Separate construction of complex object from representation such that same construction process can result in different representations

**Also Known As:** Builder

**Motivation**

- Separate object construction from business logic
- Promote readability and understandability
- Three key objects: *Director, Builder, Product*

Example: Builder to build different types of vehicles [Each has engine, tyre, etc]

# Builder Pattern: Documentation

**Applicability**

- Algorithm for creating the object must be independent
  - Different parts may make up the object
  - Need not worry about how they are put together

- Construction of different representations of the object needs to be supported

# Builder Pattern: Documentation

## Structure

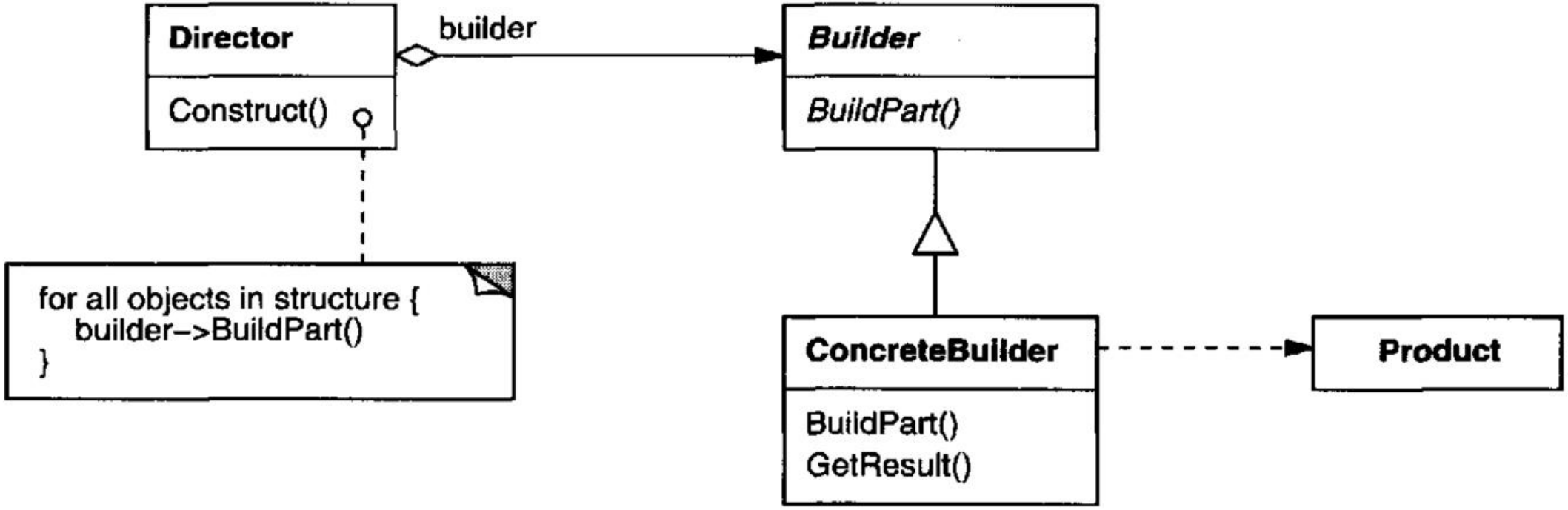



Image source: Gang of four book

# Builder Pattern: Documentation

## Participants

### Builder (StudentBuilder)

- Defines the interface for creating parts of a product object

### ConcreteBuilder (ConcreteStudentBuilder)

- Assembles the parts to create product by implementing builder interface

### Director (StudentDirector)

- Constructs an object using the builder interface

### Product (Student)

- Complex object under construction
- Includes classes that define the different parts

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# Builder Pattern: Documentation

**Consequences**

- Easily vary products internal representation
  - Director gets the abstract interface to build a product
  - All that needs to be done is to define a new kind of builder
- Isolate code for representation and constructions
  - Concrete builder contains code for building a kind of product
  - Directors can reuse builders to build different variants of product
- More control over the construction process
  - Step by step approach under directors control – Focus is on the process
- The overall code complexity increases due to multiple classes
  - Benefits in the long run

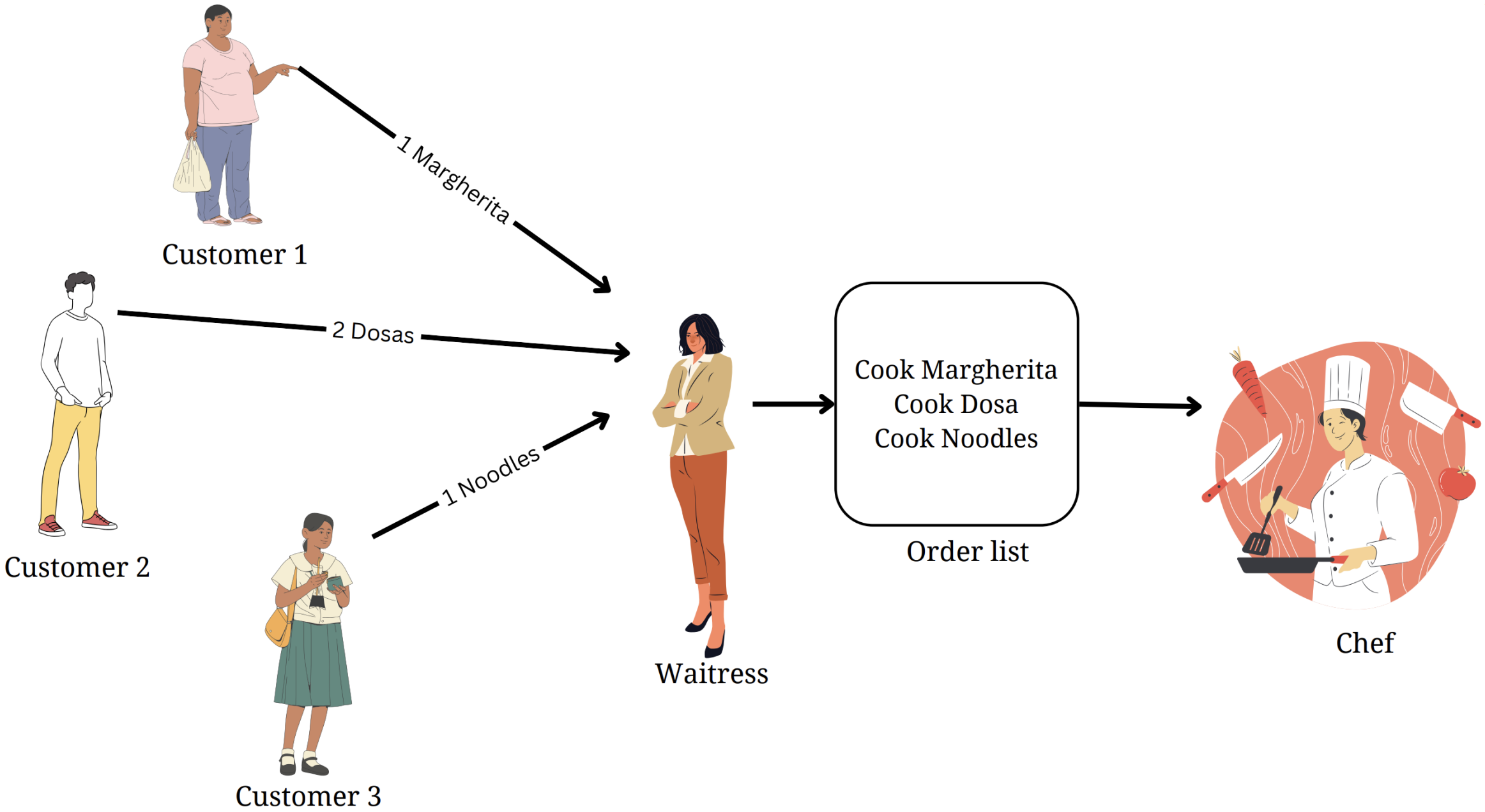
# Builder Pattern: Documentation

**Implementation**

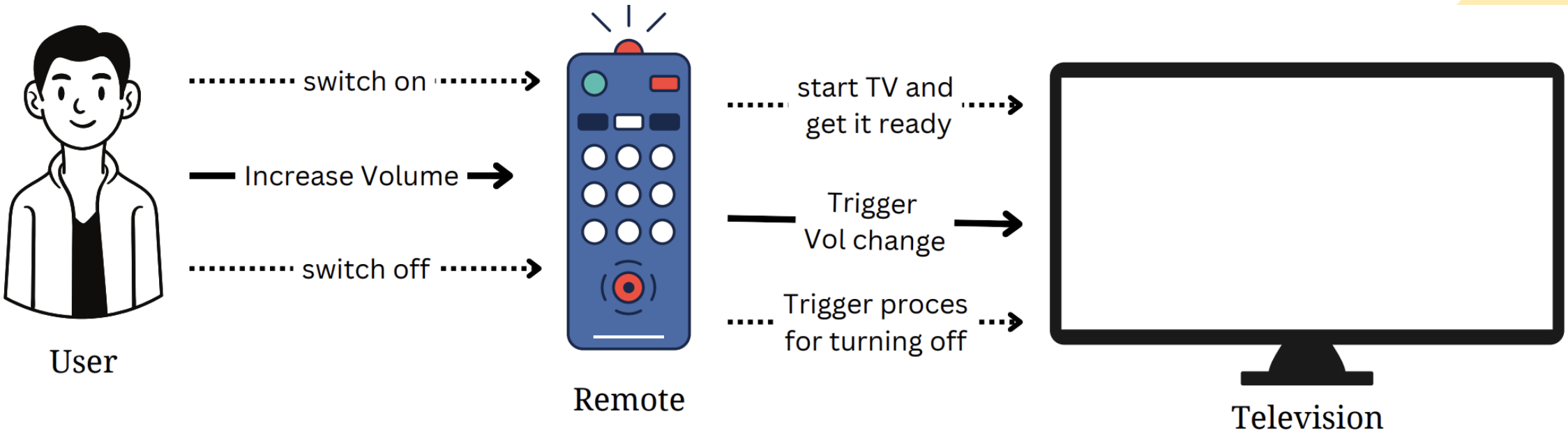
Check the source code given along: StudentRecordBuilder

# You can give a command: Command Pattern [Behavioral]

# Meet the Command Pattern!



# Meet the Command Pattern – A Scenario




Should remote know exactly how the TV work step by step?

# Meet the Command Pattern

- What if sender need not have to worry about receiver's internal implementation?
- What if some commands needs to be scheduled and executed in order at a later time?
- Sender needs to be decoupled from a receiver
- Encapsulates everything required to perform an action
  - Execution of action can happen independently

# Command Pattern: Documentation

**Intent**

Encapsulate a request as an object, allowing parameterization of clients with requests, log or queue request and support undoable operations.

**Also Known As:** Action, Transaction

**Motivation**

- Sometimes its necessary to request to objects without details about operation
- Objects can be stored and passed around -
- Five key objects: *Client, Command, Concrete Command, Invoker and Receiver*

Example: UI kits [Think about if you want to develop a button class]

# Command Pattern: Documentation

**Applicability**

- Parameterize objects by an action to perform – Callbacks in procedural
- Specify, queue, execute request at different times
- Support undo operations – Think of editors, games [Add another operation in command interface]
- Support logging changes – Manage crashes
- Sometimes an operation may be composed of primitive operations

# Command Pattern: Documentation

## Structure

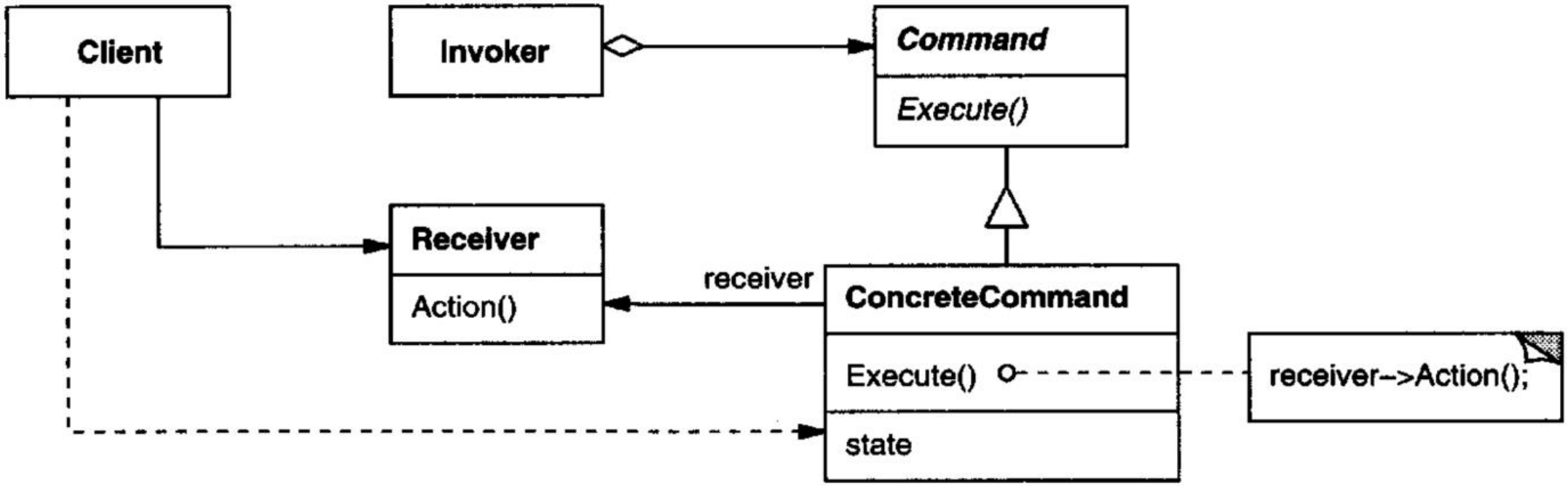



Image source: Gang of four book

# Command Pattern: Documentation

## Participants

### Command (Command.java)

- Interface for executing an operation

### ConcreteCommand (TVOnCommand, TVOffCommand,..)

- Binding between receiver object and action
- Implements the execute by invoking operations on receiver

### Receiver (Television)

- Knows how to perform the operations associated with a request

### Client (RemoteControlDemo)

- Create ConcreteCommand object and sets its receiver

### Invoker (RemoteControl)

- Calls command to execute a request



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# Command Pattern: Documentation

**Consequences**

- Decoupling client and receiver
  - Decouples invoke operation from the one that knows how to perform it
- Commands as first-class objects
  - Command can be manipulated and extended like any other object
- Composite commands can be formed
  - Commands can be composed to form a larger command
- Code complexity may increase
  - Not every time this is needed
  - Introduction of new layer between senders and receivers

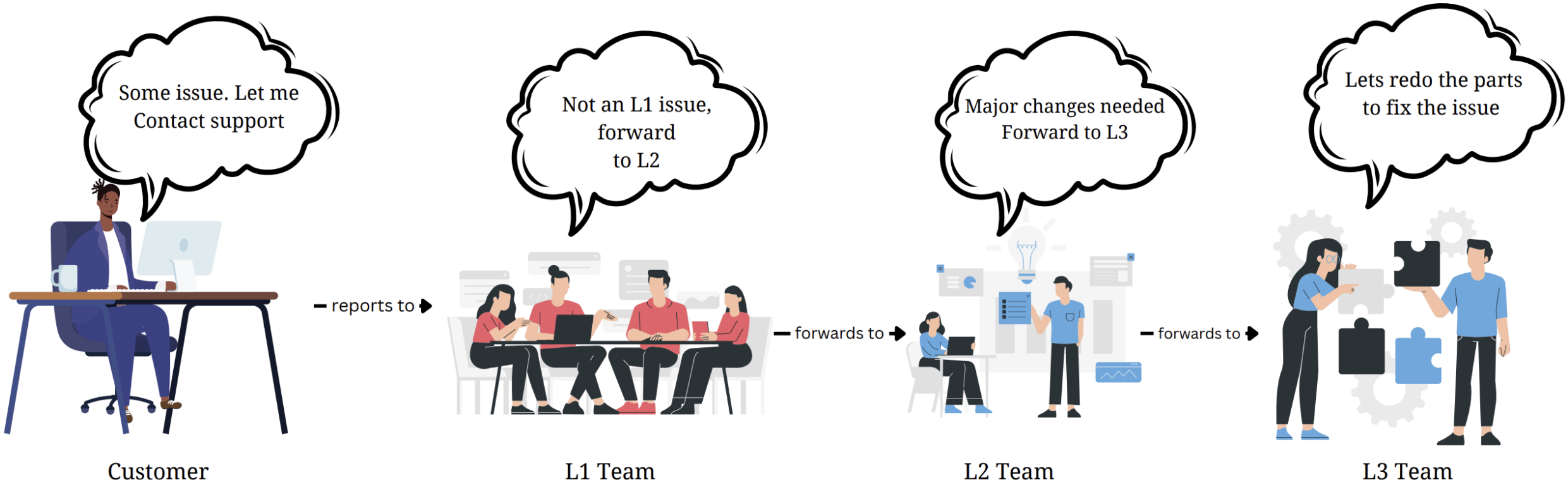
# Command Pattern: Documentation

**Implementation**

Check the source code given along: RemoteControlCommand

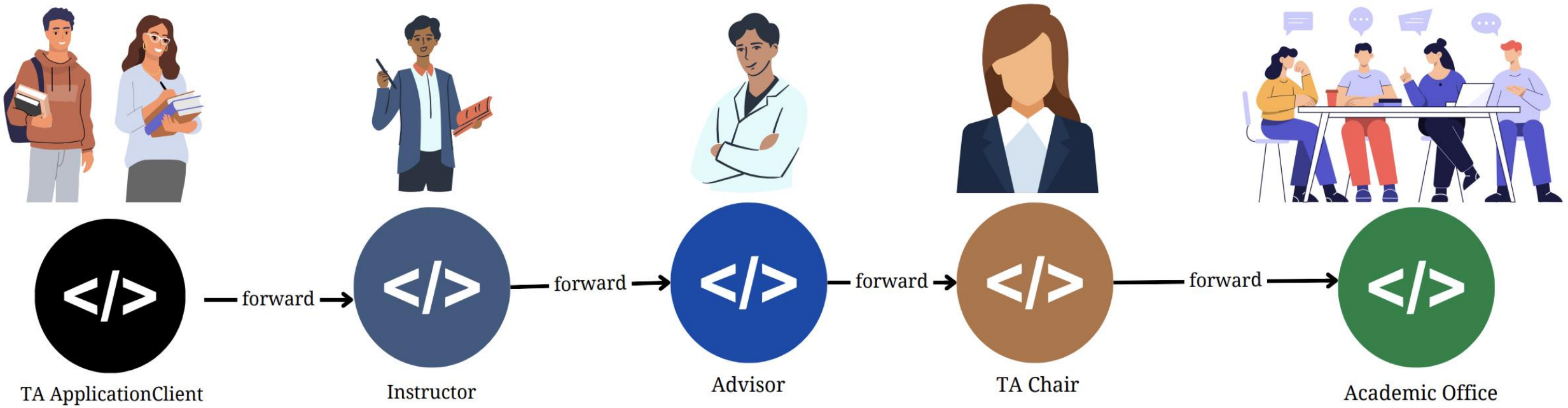
# We can pass on responsibilities: Chain of Responsibilities! [Behavioral]

# Meet the Chain of Responsibility Pattern!



# Meet the Chain of Responsibility Pattern - Motivation

TA Application Scenario




How do you implement this ?

# Meet the Chain of Responsibility Pattern

- What if one single request requires processing by multiple objects?
- What if the sender needs to be decoupled from receiver in the form of set of intermediatory objects?
- Sometimes single task may require multiple steps to process
- Each step in the process may decide if it needs to be further processed or not

# Chain of Responsibility Pattern: Documentation

**Intent**

Avoid coupling the sender of a request  to its receiver by giving more than one object a chance top handle the request. Chain the receiving objects and pass the request along the chain until one handles it

**Also Known As:** CoR, Chain of Command

**Motivation**

- Request may have to be passed along a chain
- Senders and receivers need decoupling
- Key objects: *Handler, ConcereteHandler* and *Client*

Example: Payment process in an e-commerce system

# CoR Pattern: Documentation

**Applicability**

- More than one object may handle a request and handler isn't known apriori
- Issue request to one object without specifying the receiver
- The set of objects that can handle a request should be specified dynamically

# CoR Pattern: Documentation

**Structure**

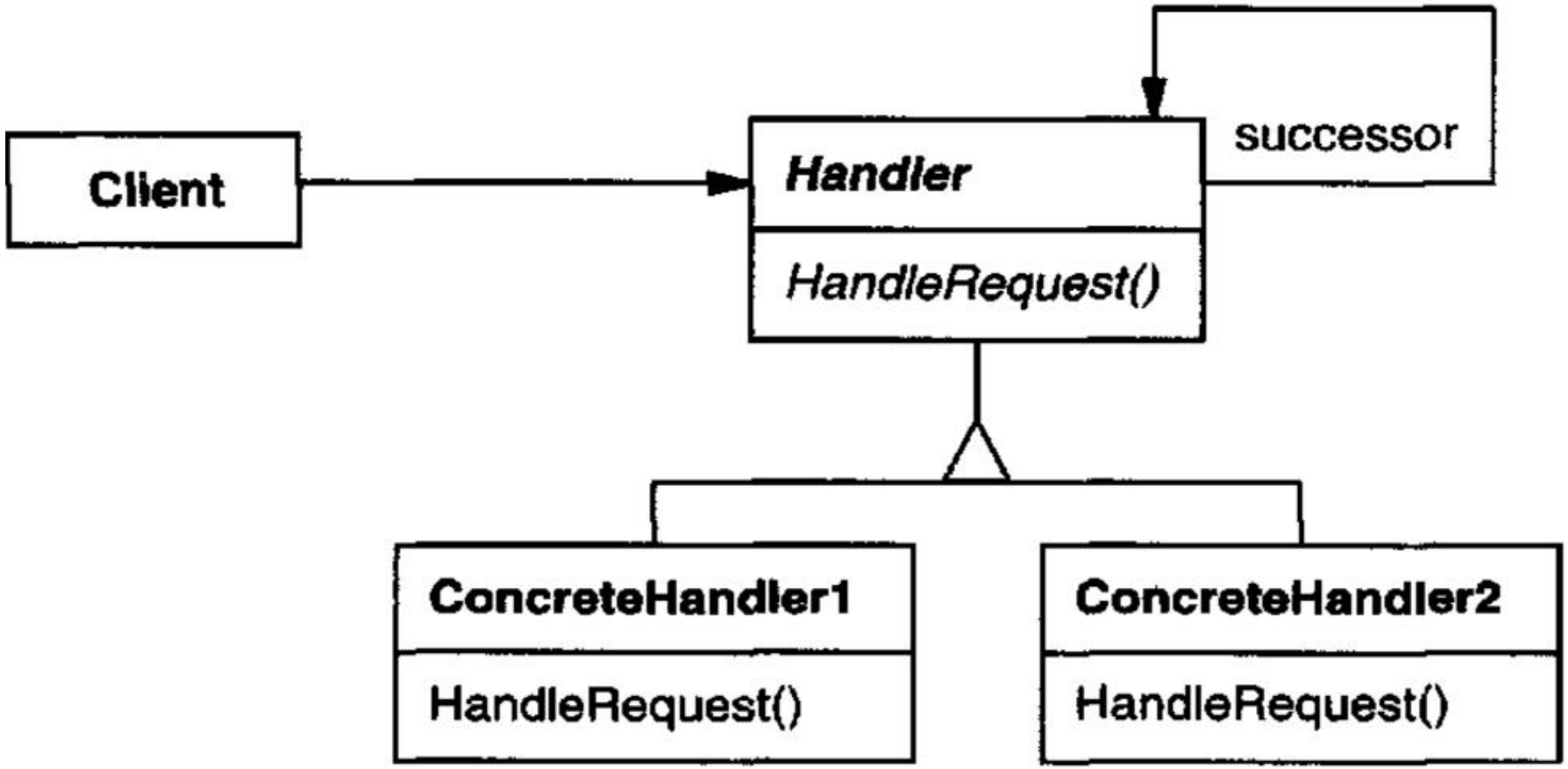



Image source: Gang of four book

# CoR Pattern: Documentation

## Participants

### Handler (ApplicationHandler)

- Defines an interface for handling requests

### ConcreteHandler (InstructorHandler)

- Handles requests its responsible for
- Can access its successor

### Client (StudentDemo)

- Initiates the request to a ConcreteHandler object on the chain

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# CoR Pattern: Documentation

**Consequences**

- Reduced Coupling
  - Object does not need to worry about which other object handles request
  - Simplifies object interactions

- Flexible assignment of responsibilities
  - Flexible distribution of responsibilities among objects
  - Responsibilities of each handler can be changed at run time (chain can be increased)

- Receipt isn't guaranteed
  - Request has no explicit receiver – No guarantee of handling
  - Request can go unhandled when chain is not configured properly

# CoR Pattern: Documentation

**Implementation**

Check the source code given along: TA-ApprovalChainOfResponsibility

# Thank You

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